

From Textual Data to Theoretical Insights: Introducing and Applying the Word-Text-Topic Extraction Approach

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Abstract

Text analysis, particularly custom dictionaries and topic modeling, has helped advance management and organization theory. Custom dictionaries involve creating word lists to quantify patterns and infer constructs, while topic modeling extracts themes from textual documents to help understand a theoretical domain. Building on these two approaches, **we propose another text analysis approach called word-text-topic extraction (WTT)**, which enhances the efficiency and relevance of text analysis for the sake of theoretical advancement. Specifically, we first identify relevant words for a researcher's theoretical area of interest using word-embedding algorithms. That step is followed by extracting text segments from the textual corpus using a collocation process. Finally, topic modeling is applied to capture themes relevant to the specific theoretical area of interest. To illustrate the WTT approach, we explored one research area needing further theory development—innovation. Using 841 CEOs' letters to shareholders, we found that our WTT approach provides nuanced features of innovation that differ across industry contexts. We guide researchers on decisions and considerations related to the WTT approach in order to facilitate its use in future studies.

Keywords

text analysis, word-embedding, collocation, topic modeling, dictionary, innovation

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Introduction

Text analysis has helped advance management and organization theory by *developing and using measures of constructs* (Short et al., 2010; Speer et al., 2022; Zachary et al., 2011), *uncovering themes connected to theoretical domains* (Croidieu & Kim, 2018; Kaplan & Vakili, 2015; Min et al., 2021; Zhang et al., 2021), and *clarifying boundary conditions* (McKenny et al., 2018; Pandey & Pandey, 2019). The growth of available large-scale data sources (e.g., extensive collections of digitized documents, archives, and/or online materials) and accessible tools (e.g., web scraping and advanced text analysis techniques) points to a bright future for continued theoretical insights derived from text analysis (D'Mello et al., 2022; Hannigan et al., 2019; Kobayashi et al., 2018; Lee et al., 2020).

Dominant text analysis approaches in management research include custom dictionaries and topic modeling. Simply put, custom dictionaries are word lists that researchers create to represent a theoretical domain. These lists are then applied to textual materials in order to quantitatively assess patterns and draw inferences about constructs (Antons et al., 2023; Duriau et al., 2007; Short et al., 2010). Topic modeling captures themes (i.e., topics) within a set of textual documents. Extracted topics allow researchers to see patterns in unstructured text data to better understand a theoretical domain (Hannigan et al., 2019; Kaplan & Vakili, 2015; Schmiedel et al., 2019; Zhang et al., 2021).

Both approaches have advantages and disadvantages. Custom dictionaries can be developed and vetted to be used by management researchers in order to reliably measure constructs from raw text. Being deductive and building on existing definitions of theoretical constructs, the custom dictionary approach may be less likely to uncover new or unexpected insights from textual data. In contrast, topic modeling takes a data-driven approach and extracts themes based on observed word cooccurrence. Being inductive, topic modeling can efficiently reveal a variety of themes in a textual dataset; but the resulting themes may not align with the researcher's interests, possibly limiting the relevance of the theoretical insights derived.

Building on these approaches, we propose a new text analysis process called word-text-topic extraction (WTT), which provides an efficient and focused way to explore a theoretical domain in large textual datasets. WTT combines elements of both custom dictionaries and topic modeling and offers a more nuanced approach to managing diverse and large-scale textual data. In the following sections, we discuss these approaches to text analysis and how the WTT complements and builds them. After clarifying the decisions related to undertaking this approach, we demonstrate its use through our investigation of one theoretical domain—innovation. For our empirical application of WTT, we gathered textual data from 4 years of CEOs' letters to shareholders in tech- and nontech-related industries. We found that our approach provides nuanced features of innovation that are differentiated across industry contexts. We clarify the decisions and considerations when undertaking this approach so that management researchers can use it to explore theoretical areas and augment theoretical understanding.

Literature Review

In this section, we discuss more in depth the two approaches to text analysis—custom dictionaries and topic modeling—and their use in management research.

Custom Dictionaries

Well-vetted, **predefined dictionaries**, such as Linguistic Inquiry and Word Count and DICTION, have been widely used in management research (e.g., Pfarrer et al., 2010; Short & Palmer, 2008; Zavyalova et al., 2012). These dictionaries are built on extensive research and empirical testing, ensuring that they accurately categorize and quantify text related to various constructs, such as

positive emotions, negative emotions, cognitive terms, and motion (Hickman et al., 2022; Short & Palmer, 2008). Using these dictionaries allows researchers to reliably analyze diverse types and sources of textual data (e.g., CEOs' letters to shareholders, social media posts, news articles, and organizations' mission statements). However, these dictionaries are limited in their ability to capture a wide variety of constructs in management research (Short & Palmer, 2008).

Given the need to develop dictionaries for management and organizational constructs, researchers have used computer-aided text analysis to develop custom dictionaries (e.g., McKenny et al., 2018; Zachary et al., 2011).¹ Short et al. (2010) clearly outlined the process for creating a custom dictionary to measure constructs of interest with an emphasis on ensuring rigorous construct validity. Researchers have created custom dictionaries to measure various management and organizational constructs—for example, *digital orientation* (Kindermann et al., 2021), *entrepreneurial orientation* (McKenny et al., 2018; Short et al., 2010), *social and economic value orientation* (Moss et al., 2018), and *market orientation* (Zachary et al., 2011). When applied to textual materials, these dictionaries provide quantitative measurements of the extent a construct is found in a textual dataset (e.g., frequencies of dictionary words in the document). These quantitative measures can be used to assess impact on outcomes, such as firm performance and organizational outcome, or to compare with other constructs, such as downsizing, emotions, and new product introduction (Duriau et al., 2007). Researchers have applied these custom dictionaries to various types of textual data (e.g., CEOs' letters to shareholders, social media posts, news articles, and narratives). This approach allows us to extract insights into various organizational contexts while avoiding the traditional challenges encountered in management studies—for example, difficulties in gathering survey responses, individuals' reluctance to participate in research, and fatigue from coding textual data (Reid et al., 2023).

Custom dictionaries provide a relevant portrayal of a construct in a textual dataset, but creating these dictionaries is very time-consuming. According to Krippendorff (2004), "Most content analyses would benefit from the construction of special purpose dictionaries, but developing a dictionary from scratch can be a formidable task" (p. 287, emphasis added). The process of creating custom dictionaries typically involves recruiting subject-matter experts to vet an initial word list derived from an extensive literature review and synonym searches. Once the dictionary is vetted, word counts are typically used to reflect how much of the construct is found in the textual data. After the dictionary application, leftover words require additional expert review to either integrate them into existing dictionaries or identify their connection to a construct's new dimensions (Short et al., 2010). Moreover, researchers must ensure that words are used in the way in which they are intended in the dictionary; therefore, an intensive keyword-in-context (KWIC) review or manual coding of a subsample is required (e.g., Palrecha et al., 2012; Short et al., 2018). These steps in creating a custom dictionary are time-consuming tasks, which researchers must undertake before conducting construct validity measures—for example, content, external, discriminant, and predictive measures (Short et al., 2010).

Therefore, custom dictionaries provide a relevant set of words closely associated with a theoretical construct. Uncovering unexpected, previously unknown insights into textual data is not the primary objective of the custom dictionary approach. Instead, because these dictionaries are developed deductively by using preexisting definitions, this approach is primarily intended to reliably measure theoretical constructs in the textual data to test theories and extend understandings across different contexts.

Topic Modeling

Topic modeling is an approach to text analysis that has been increasingly adopted in management research. It uses algorithms to extract topics in a large set of documents (i.e., a corpus). Topic modeling can identify **themes** (i.e., groupings of words) based on statistical patterns, such as the

cooccurrence of words in textual data (DiMaggio et al., 2013; Schmiedel et al., 2019; Turney & Pantel, 2010). It can be rapidly applied to identify topics or themes from a textual dataset (Hannigan et al., 2019; Kobayashi et al., 2018; Schmiedel et al., 2019), thus enabling the discovery of new topics without requiring extensive human guidance.

Topic modeling has been used to capture various themes from textual data. For instance, management researchers have used it to identify theoretical concepts, such as *novel innovations* (Kaplan & Vakili, 2015), *strategic distinctiveness* (Haans, 2019), *strategic focus* (Doldor et al., 2019), *institutional logics* (Jha & Beckman, 2017), and *legitimation mechanisms* (Croidieu & Kim, 2018). Once the topics are extracted from textual data, researchers can assess these results and their connection to a theoretical area. Results can also be used to identify how topic patterns change over time and across contexts, to assess the importance of topics, and to understand topic connections with external factors (e.g., Antons et al., 2019; Croidieu & Kim, 2018; Kaplan & Vakili, 2015). Topic modeling has grown in popularity because it allows management researchers to rapidly discover themes in large textual datasets (Hannigan et al., 2019).

A shortcoming of topic modeling is that resulting topics may not necessarily be relevant to a researcher's theoretical area of interest. As an unsupervised machine learning technique, topic modeling identifies topics as words that tend to be found together in textual data (Marshall et al., 2023). In current management research that employs topic modeling, extracting over 100 topics from large-scale textual data is not unusual (e.g., Banks et al., 2019; Croidieu & Kim, 2018; Kaplan & Vakili, 2015). Researchers must sort through results to find a subset of topics relevant to their theoretical domain of interest. Recognizing both the strengths and the limitations of the above approaches, we developed an integrating process building on the custom dictionary and topic modeling approaches, as explained next.

The WTT Extraction Approach

Our three-step WTT approach provides researchers with an efficient and effective way to capture themes relevant to their theoretical areas of interest. The “W” represents the *word extraction* step. The goal is to obtain a comprehensive word list related to a theoretical concept of interest, similar to the goal of creating custom dictionaries. Unlike the traditional practice of curating custom dictionaries, this step is accomplished through word-embedding algorithms. This step's output is a list of words similar to the theoretical concept of interest with respect to the context in which they are used. The first “T” represents the *text extraction* step, which involves applying the word list (from the W step) to researcher-collected documents and extracting text around each word, a process known as *collocation* (Gephart, 1997). These collocated words make a more focused corpus and are used as input in the next step. The second “T” is the *topic extraction* step, in which researchers apply topic modeling to the extracted text segments in order to identify themes.

The remainder of this section provides details and key decisions for each step. Figure 1 illustrates the WTT approach's flow, and Table 1 lists the key decisions and considerations for each step.

Step 1: Word Extraction (W)—Finding Similar Words Via Word-Embedding Algorithms

In the word extraction step, **researchers develop a comprehensive word list related to their theoretical area of interest**. They begin with a small set of words (referred to as *focal words*) closely related to the concept of interest. Then they **employ a word-embedding algorithm to expand the search for words similar to the focal words**; the resulting list of words is called *hit words*. Collectively, we refer to the focal and hit words as *search words*.

Word-embedding algorithms help researchers efficiently find keywords similar to the initial list of focal words. These algorithms convert words into an array of numbers (known as a *vector*), such that

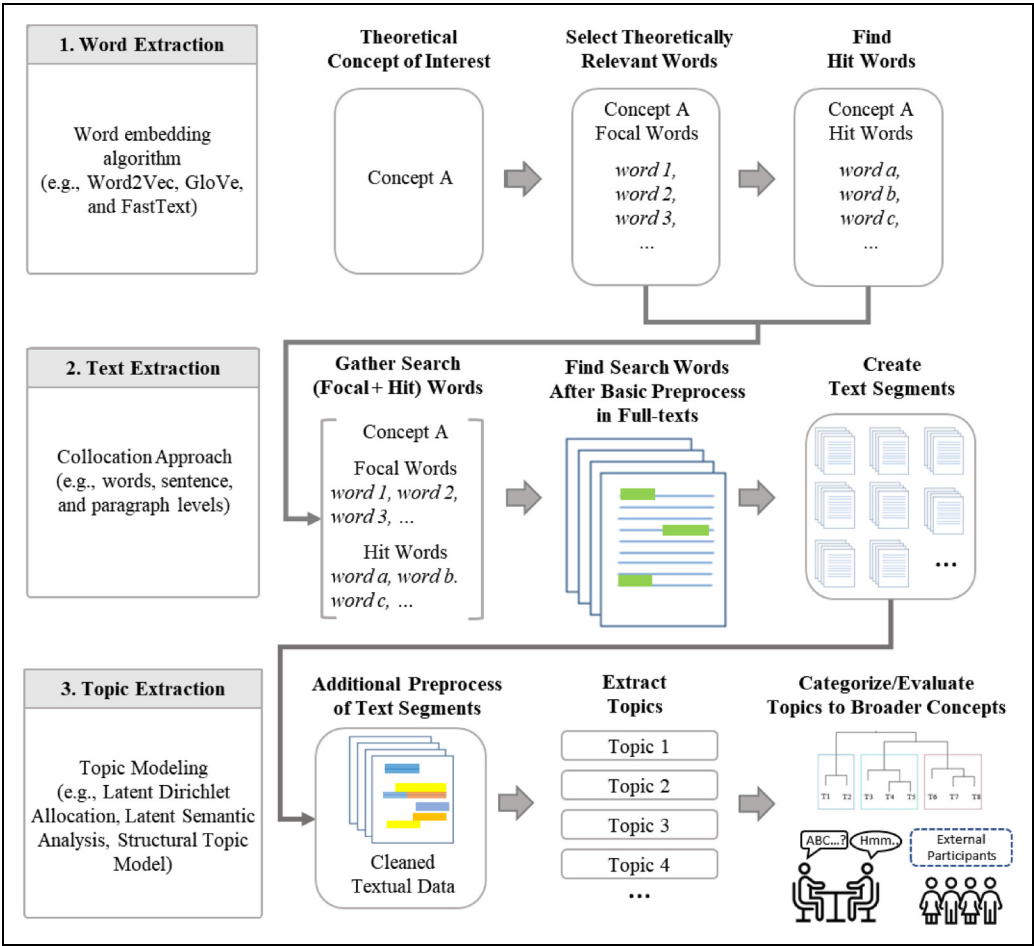


Figure 1. Summary of the WTT approach to understand further a theoretical concept.
Note: WTT = word-text-topic extraction.

words with similar meanings have a higher similarity score (usually measured with the cosine similarity). Word-embedding algorithms are designed to help identify the semantic relatedness among words based on occurrences in large quantities of texts. Such semantic relatedness is trained by examining the similarity of surrounding words known as *embedding* (Mikolov, Chen, et al., 2013; Mikolov, Sutskever, et al., 2013). For example, an algorithm would identify the words “sustainability” and “environmental” as semantically related if they are frequently used in similar phrases like “meet the *sustainability* challenges” and “meet the *environmental* challenges” or “responsible *sustainability* guidance” and “responsible *environmental* guidance.”

Decisions and Considerations. Four major decisions and considerations are related to the word-extraction step (see Table 1). First, researchers should identify the initial focal words related to the core theoretical concept found in published research articles, check words used in surveys from published papers, and review the final list of words with other researchers or experts in their research domain.

Second, researchers should decide which word-embedding algorithms to use in order to generate hit words. In the quickly developing field of representation machine learning, popular methods

Table 1. Researcher Decisions for the Word-Text-Topic Extraction (WTT) Approach.

Steps	Decisions	Considerations
Word extraction (W)	1. Selecting theoretically relevant words (focal words)	To identify the appropriate initial keywords: 1. Consider the published definition of a theoretical concept and the words used in surveys or custom dictionaries; 2. Develop keywords and find their different forms (e.g., nouns, verbs, and adjectives) that are related to the concept; and 3. Refine word lists through discussion with researchers and experts.
	2. Selecting an algorithm	Choose a word-embedding algorithm based on the study's needs, the nature of the task and data, and the availability of a suitable pretrained model. The following are examples of word-embedding algorithms: • Word2Vec (https://code.google.com/archive/p/word2vec/) learns from data to understand word relationships based on their cooccurrence in a text. By focusing on local context (nearby words), it captures specific relationships between words in similar contexts (Mikolov, Chen, et al., 2013; Mikolov, Sutskever, et al., 2013). • GloVe (https://nlp.stanford.edu/projects/glove/) learns from the cooccurrence of words as does Word2Vec, but it considers the entire corpus. Thus, it can identify broader relationships between words (Pennington et al., 2014). • FastText (https://ai.facebook.com/tools/fasttext/), an extension of Word2Vec and GloVe, breaks words into character n-grams (e.g., "running" becomes "run, unn, nni, nin, ing"), enabling it to understand related words, misspelled words, compound words, and unknown words not included in the training data (Bojanowski et al., 2017).
	3. Selecting the model: pretrained or custom-trained	Decide whether to use a pretrained model or to train on a researcher-created dataset. Consider the pretrained model's data source (e.g., news or social media) and the amount of data collected. • Use a pretrained model if the data source is similar to the researcher's created dataset and the amount of data for the research is limited. • Use a custom-trained model if the researcher's textual dataset is unique and a large amount of data is collected.
	4. Evaluating the search words	There are no absolute cut-off (quantitative) criteria, but researchers can use cosine similarity scores to help guide their decisions (i.e., balancing the quantity and relevance of the hit words). The selection and finalization of word lists may also require the individual interpretations of researchers and consensus via discussion among themselves.

(continued)

Table 1. (Continued).

Steps	Decisions	Considerations
Text extraction (T)	1. Basic preprocessing decisions	Preprocessing text data • Lowercase transformation can be required because uppercase and lowercase words are stored differently in computer language; therefore, use it as a mandatory step unless there are specific reasons not to, such as when key terms have different meanings depending on capitalization. • Removing numbers or punctuation can be required to reduce noise or nonessential information; thus, remove them unless they provide research-related information (e.g., individual differences in writing style). Also, researchers should avoid removing punctuation if they aim to capture sentence or paragraph-level textual data. • Researchers are free to perform these or other preprocessing tasks (e.g., stopwords or spell correction) as needed.
	2. Selecting text segments: word, sentence, and paragraph levels	Using search words from the word extraction (W) step, retrieve text segments from documents that provide details about a theoretical concept of interest. • Depending on the researcher's needs, various sizes of text segments can be extracted—for example, the number of words on either side of keywords, sentence level, and paragraph level. • Evaluate sample text segments to ensure they contain sufficient information on a concept. • Avoid double counting (word levels); searched keywords can be close enough within sentences that extracted words can overlap.
Topic extraction (T)	1. Additional preprocessing decisions	Further preprocessing text data can be performed to enhance data quality and identify key patterns rather than noise (e.g., Kobayashi et al., 2018; Schmiedel et al., 2019). • Removing stop words, uninformative and high-frequency words (e.g., <i>a</i>, <i>or</i>, <i>an</i>), can be required to reduce noise in topic modeling results. • Although lists of stop words are available in text analysis tools, researchers can also create custom stop words to exclude meaningless words, both frequently occurring words (e.g., names of companies and products) and those with low frequency (e.g., technical or specialized terms). • Other preprocessing tasks (e.g., stemmer, lemmatizer, or spell correction) can be performed based on the researchers' needs.
	2. Extracting topics	There are no clear guidelines for choosing the right number of topics based on textual data. • Use quantitative measures (e.g., coherence or

(continued)

Table 1. (continued)

Steps	Decisions	Considerations
		log-likelihood scores) as references (e.g., Chang et al., 2009; Mimno et al., 2011).
		• Extract different numbers of topics (e.g., 25, 50, 100, etc), identify related topics using a clustering analysis (i.e., Ward's method), and evaluate outcomes with other researchers and experts (e.g., Croidieu & Kim, 2018; Kaplan & Vakili, 2015).
	3. Capturing reliability and face validity	Prior management studies discussed capturing reliability and face validity: • Label topics individually and collectively with researchers and/or experts (e.g., Antons et al., 2019; Kaplan & Vakili, 2015). • Evaluate topics and words to enhance their reliability and face validity by inviting external participants (Chang et al., 2009; Kjell et al., 2019).

include Word2Vec, Global Vectors (GloVe), and FastText. Each word-embedding algorithm employs a unique computational model (Kumar, 2021) and differs slightly from others in terms of accuracy, understanding of words, and relationships with other words (see Table 1, the second section “Selecting the Algorithm” under “Word Extraction (W)” for more information). For instance, Word2Vec identifies relationships between words based on the similarity of their neighboring words within a document (Mikolov, Chen, et al., 2013). In contrast, GloVe examines word pairings throughout an entire set of documents, finding broader associations between words (Pennington et al., 2014). FastText breaks words into character n-grams (i.e., the character 3-gram for *running* would include *run*, *unn*, *nni*, *nin*, and *ing*), allowing it to handle misspellings, related words, and even unknown words (Bojanowski et al., 2017). There are no hard-and-fast rules for selecting an algorithm. Instead, the choice of a word-embedding algorithm depends on a study’s specific requirements, including the nature of the task and data.

Third, along with selecting the algorithm, researchers can choose whether to use *pretrained models*, which are ready-to-use algorithms trained on large datasets, or *custom models*, which are trained on a researcher’s unique dataset. To use a pretrained model, researchers can load and infer word vectors without going through the model training process because the word-embedding algorithm (e.g., Word2Vec, GloVe, or FastText) is already trained on a large set of textual data (e.g., online news, social media, or Wikipedia). Freely available and time-efficient, a pretrained model is best when the pretrained model’s data is similar to the type of data the researcher has collected for investigating a theoretical construct. For instance, if an algorithm is trained on public news accounts, the resulting words provide a good fit for a researcher using media articles on companies. Alternatively, researchers can use their own textual data to train a custom model and identify hit words. This option is best suited for researchers who have a very large textual dataset. If the researcher has developed a large corpus, this customized approach could yield more accurate and specific hit words compared to a pretrained model.

Fourth, a word-embedding algorithm generates a list of words (i.e., hit words) related to focal words reflecting an area of theoretical interest. The hit words must be evaluated for their pertinence to the theoretical concept. Researchers/experts may use quantitative approaches (e.g., cosine

similarity) and/or work closely together to interpret words/phrases related to a concept and finalize word lists reflecting that concept (Bojanowski et al., 2017; Pennington et al., 2014; Short et al., 2010). We offer some advice on this issue in our example study.

Step 2: Text Extraction (T)—Text Parsing Using Search Words

Once the search words (i.e., hit and focal words associated with an area of theoretical interest) are created, researchers can use them to **extract text segments** (i.e., text around each search word) from the raw documents. Referred to as **collocation**, this extraction process retrieves text, or “mini-texts,” around search words (Gephart, 1997, p. 594). In other words, this “T” process parses the researcher’s documents to include only those words in close proximity to the search words. These text segments provide “evidence of ‘meaningful linkages’ among keywords in subjects’ discourse as well as to establish linkages among these keywords and the concepts” (Gephart, 1997, p. 587). By using text segments instead of the entire corpus, researchers can obtain more relevant information about an area of theoretical interest while avoiding noise resulting from using the entire text (Gephart, 1997; Pollach, 2012).

Decisions and Considerations. In text extraction, two decisions are involved: (a) how the researcher will preprocess the textual data and (b) how much of the text surrounding the search words should be retained. First, researchers should consider undertaking basic preprocessing tasks, such as lowercase transformation (i.e., format removal), punctuation removal, and number removal (e.g., Hickman et al., 2022; Kobayashi et al., 2018). Preprocessing ensures that words are captured based on their spelling regardless of letter case.²

Second, researchers must consider how much of the text surrounding the search words should be extracted from textual data. The length of text extracted around each search word can vary depending on the textual data’s characteristics. Some documents may straightforwardly discuss a single theme (e.g., short social media posts, short answers in a survey, or responses in a legal proceeding), whereas others may have broad discussions containing many themes (e.g., CEOs’ letters to shareholders or interview transcripts). Researchers must experiment with various ways of parsing textual data (e.g., phrases, sentences, and paragraphs) and determine how the amount of text extracted affects the next step—topic modeling—and its results. Unfortunately, there are no hard-and-fast rules to make this assessment; therefore, research judgment is required. In our example, we provide guidance on this issue.

Step 3: Topic Extraction (T)—Extracting Themes Using Topic Modeling

At this point, the researcher’s dataset consists of extracted text segments to which researchers can apply topic modeling. The topic modeling process captures groups of words (i.e., topics) frequently cooccurring in the same unit (e.g., sentence, paragraph, or document) to convey semantic meanings (Kobayashi et al., 2018; Schmiedel et al., 2019). These topics may be used to represent theoretical concepts after researchers and theoretical domain experts collectively interpret and thoroughly validate them (e.g., Croidieu & Kim, 2018; Haans, 2019; Hannigan et al., 2019; Lu & Chesbrough, 2022; Zhang et al., 2021). The resulting topics can be used for further qualitative and quantitative analysis. Topics can also be further grouped based on the researcher’s interpretations and/or statistical clustering analysis, such as hierarchical and k-means clustering (Kobayashi et al., 2018).

Decisions and Considerations. Topic modeling involves three broad decisions: (a) how to preprocess the textual data (i.e., additional data cleaning from the previous T step), (b) how to extract topics, and (c) how to assess the quality of the results.

First, before topic modeling is conducted, additional preprocessing tasks may be required to enhance data quality (Kobayashi et al., 2018; Schmiedel et al., 2019). For instance, researchers

may want to remove stop words (e.g., conjunctions and prepositions), which frequently appear in text while not carrying inherent meanings. Additionally, they may remove terms with frequently appearing words that have no meaning (e.g., specific company and product names) and delete low-frequency terms (e.g., technical and specialized terms) that do not offer robustness for statistical modeling. To normalize text by stacking different forms of the same words, researchers may also conduct stemming (reducing words to their stems: e.g., from “changing” to “chang”) or lemmatization (reducing words to their dictionary form: e.g., from “changing” to “change”). Additional options (e.g., emoticon recognition, expansion of contractions and abbreviations, and spelling correction) may be considered for fine-grained text cleaning. In summary, preprocessing tasks must be relevant to the study (Hickman et al., 2022).

Second, researchers should determine how many topics are appropriate given their textual data. The question of how to capture the right number of topics is an ongoing debate (Giorgi et al., 2019). The appropriate number of topics is determined by the content’s diversity relative to the corpus characteristic (e.g., document size), the level of intended granularity, and the ability to interpret the topics. Computer scientists use quantitative approaches (e.g., coherence and log-likelihood scores) to identify the optimal number of topics (e.g., Chang et al., 2009; Mimno et al., 2011). However, Chang et al. (2009) found that researchers’ interpretations of topics did not align with results from these quantitative measures. Therefore, management researchers have used qualitative approaches to determine the appropriate number of topics (e.g., Croidieu & Kim, 2018; Hannigan et al., 2019). The general principle is based on interpretability. Researchers must review the output from different numbers of topics (e.g., 25, 50, 100, etc.) to determine at what level the resultant topics are the most illuminating to their theoretical interest (e.g., Huang et al., 2018; Shang & Luo, 2022).

Finally, once identified, the resulting topics can be labeled and categorized based on their connection to the theoretical concept of interest (Banks et al., 2019; Croidieu & Kim, 2018). For instance, Croidieu and Kim (2018) theorized mechanisms of lay-expertise legitimation by studying the U.S. wireless radio broadcasting sector. Using various historical textual data (e.g., popular media, conferences, regulations, and annual reports) related to U.S. amateur radio operators, the authors captured 98 topics, labeled all the topics, and broadly categorized them into four legitimation mechanisms based on researchers’ interpretations. Schmiedel et al. (2019) stated, “To ensure the reliability ... researchers should use multiple coders and ... reach consensus about the meaning of topics. Regarding face validity, researchers need to carefully examine the relevance of the identified topics for their specific study context” (p. 950). Thus, after researchers initially discuss, label, and categorize topics, they can invite external experts to interpret the topics (Chang et al., 2009; Kjell et al., 2019; Lu & Chesbrough, 2022).

Summary of the Three Approaches: Custom Dictionary, Topic Modeling, and WTT

The following is a summary and comparison of the three approaches—custom dictionary, topic modeling, and WTT approaches—based on various criteria, including theoretical concept identification, application, validation, and challenges (see Table 2). First, related to theoretical concept identification, a custom dictionary approach relies heavily on researchers’ and experts’ in-depth knowledge of a theoretical domain. The WTT and topic modeling approaches are more data-driven; researchers must sufficiently understand a theoretical domain to interpret and make sense of the results. Second, applications of the different approaches vary. The custom dictionary approach focuses on measuring specific constructs, WTT captures the prevalence of themes within selected textual materials, and topic modeling discovers fresh insights into the entire documents. Third, the custom dictionary approach primarily relies on various types of validity, such as content, external, discriminant, and predictive validity (Short et al., 2010). The WTT approach uses both statistical methods

(e.g., cosine similarity) and experts'/researchers' evaluations of collocation and topic modeling results. Topic modeling also relies on researchers'/experts' interpretations (e.g., Hannigan et al., 2019; Schmiedel et al., 2019). Finally, in terms of challenges, a custom dictionary can limit exploration of other theoretical insights, and topic modeling may not find insights related to research interests. One WTT challenge is finding the right balance between selecting an appropriate number of search words and determining the scope of extraction (i.e., words, sentences, or paragraphs around the search words). If the extraction around the search words is too narrow, researchers may lose key theoretical insights. If it is too broad, the resulting topics will contain unrelated information.

Illustrating the WTT Approach: Innovation in Different Contexts

We illustrate the WTT process by applying it to the innovation theoretical area. Innovation is an area of management and organizational research that needs further theoretical insights given mixed findings about this construct across different contexts (Brown & Eisenhardt, 1997; Damanpour, 1991; Damanpour et al., 2009). One reason for the mixed findings could be attributed to how innovation has been measured (Adams et al., 2006; Nelson et al., 2014). For instance, Armbruster et al. (2008) argued that a general measure of innovation may be insufficient because this theoretical concept is multidimensional. Damanpour (1991) noted that different industries, opportunities, and threats are examples of environmental factors influencing innovation. Thus, our study included data from two contexts: high-tech and nonhigh-tech industries.

To measure innovation, management researchers have frequently used a firm's quantitative information, such as *the number of patents and R&D expenditures* (e.g., Hull & Rothenberg, 2008; Spencer, 2003) or *responses to innovation scales on a survey* (e.g., Govindarajan & Kopalle, 2006; Knight, 1997). More recently, management researchers explored the possibility of using text analysis to develop innovation-related measurements due to the availability of advanced text analysis tools (Antons et al., 2020). For instance, Short et al. (2010) created a custom dictionary for innovativeness, which has been widely adopted in subsequent studies (e.g., Boling et al., 2016; Keil et al., 2017). This dictionary captures a general understanding of innovation rather than specific dimensions. Bellstam et al. (2021) used topic modeling to measure innovation using stock analyst reports, but they found that only one topic out of 15 was related to innovation. Therefore, we chose to apply our WTT approach to the innovation theoretical domain, and we compared WTT results with both custom dictionary and topic modeling approaches, using the same dataset.

Data Collection

To conduct our investigation, we created a textual dataset from letters to shareholders, in which CEOs discussed important aspects of their business such as innovations (Pandey & Pandey, 2019). We collected 4 years of annual reports (2012 to 2015) from 254 Russell 2000 companies, which were chosen because smaller firms are considered more innovative than larger ones (e.g., Damanpour, 1991) and discussion of innovation should be salient in letters to shareholders. Furthermore, scholars have noted that innovation differs by industry (e.g., Brown & Eisenhardt, 1997; Eisenhardt & Schoonhoven, 1996). Therefore, the companies were chosen from either high-tech or nonhigh-tech industries,³ and 841 letters⁴ were found for these companies over the 4-year span. We then used this textual dataset to compare how innovation was discussed and how it differed across these two contexts (Hall & Vopel, 1996; Kile & Phillips, 2009). Our decisions related to each step and the outcomes are discussed below.

Table 2. Comparison of the Three Text Analysis Approaches.

Three Approaches	Custom Dictionary Approach	WTT Approach	Topic Modeling Approach
Theoretical concept identification	Heavily depends on the researchers' and experts' in-depth knowledge of the theoretical concept.	- Depends on data-driven approaches. - Requires enough understanding of the theoretical concept from researchers/experts.	- Heavily depends on data-driven approaches. - Requires enough understanding of the theoretical concept from researchers/experts.
Application	Measure for a specific construct.	Prevalence of concepts/topics in a particular theoretical domain.	Prevalence of concepts/topics in documents.
Validation	Evaluate various types of validity, such as content, external, discriminant, and predictive.	- Statistical methods for word-embedding results (e.g., cosine similarity). - Researchers' evaluations of different collocation results. - Evaluated through researchers' and/or experts' topic interpretations (e.g., interrater reliability).	Evaluate based on researchers' and/or experts' topic interpretations (e.g., interrater reliability).
Challenges	- Highly relevant to a construct, so the approach can be difficult for exploring other theoretical ideas related to a construct. - Substantial time is required to develop a custom dictionary.	Uncovers multiple themes related to a theoretical concept, but it requires a careful balance during word and text extraction steps: too narrow a scope may miss important insights, while too broad a scope may include unrelated information.	Uncovers a multitude of themes inductively so that researchers may be unable to find themes or insights related to their research interest.

Note: Regarding topic modeling results' validation, while coherence measures in topic modeling provide a statistical evaluation of semantic similarity, studies have shown that these measures do not always align with researchers' interpretations of the topics (Chang et al., 2009; DiMaggio et al., 2013). Thus, the emphasis has shifted toward ensuring topic interpretability for validation (e.g., Huang et al., 2018; Kaplan & Vakili, 2015; Shang & Luo, 2022).

Demonstrating the WTT Approach

Word Extraction. We started by identifying our focal words. Because “the field of innovation is very broad” (Damanpour, 1991, p. 556), we developed broad focal words to explore different aspects of innovation from prior research rather than focusing on a specific type of innovation (e.g., product, service, process, or business model innovations). To generate hit words, we used six broad focal words: *innovation*, *innovations*, *innovate*, *innovates*, *innovative*, and *innovating*. To expand the word list, we selected the Word2Vec algorithm because of its efficiency and notable effectiveness in identifying semantic relationships. It has also been used in management research (Guo et al., 2021; Lee et al., 2020). This step is explained in detail with the Python/R code in Appendix A.

To use the algorithm, we took a pretrained approach because our textual dataset was not large enough for custom processing. The pretrained approach used Google News, which contains three

million English words from various contexts (Bolukbasi et al., 2016; Mikolov, Sutskever, et al., 2013). The Google News corpus contains mostly standard English and is similar to our corpus (CEOs' letters to shareholders) in terms of word usage (i.e., a mix of formal and conversational language). Because the Google Word2Vec model was robustly trained with this extremely large dataset (i.e., the Google News corpus), it is comprehensive in identifying hit words.⁵

As seen in Appendix A (Python/R code), we undertook several processes using Google's pretrained Word2Vec model to obtain the results. We (a) loaded a pretrained model (Word2Vec-google-news-300), (b) obtained terms most similar to our innovation focal words, and (c) used a cosine similarity score (i.e., a cosine coefficient) to determine a cut-off for our focal words' hit words.⁶ The cosine similarity scores ranged from 0 to 1, with 1 indicating complete similarity and 0 indicating no similarity.

Figure 2 shows how many words were captured from our shareholder letters depending on cosine similarity scores. We noticed a significant increase in hit words at a cosine similarity score below 0.5. Therefore, we set our threshold at 0.5, a midpoint between *no similarity* (0) and *complete similarity* (1). This threshold allowed us to avoid capturing too many tangential (slightly related) words while capturing as many relevant words as possible.

In Table 3, we present our Word2Vec results, including a list of hit words and phrases related to the six innovation-focal words. Although the list of hit words contains obvious noise, such as website and firm names that are not directly related to innovation, we retained them without making adjustments. This decision was made to remove the need for researchers to spend significant time making detailed judgment calls. Also, the impact on our WTT approach's results was expected to be minimal.

Text Extraction. Using the innovation search words developed from the word extraction (W) step, we undertook a collocation process with our textual dataset of CEO letters. We selected four words—before and after each search word in our textual dataset—to isolate and capture how the CEOs discussed innovation in their letters.⁷ We also allowed the extraction process to select words outside the sentence where the search words were found because ideas are related between sentences in well-crafted letters to shareholders. Both collocation decisions are similar to Gephart's approach (1997).

Extracting text from our letters to shareholders involved three steps (see Appendix B with Python/R code). First, we cleaned the data in the letters to shareholders by changing all text to lowercase letters⁸ and removing punctuation and numbers as discussed previously. Second, using a KWIC function,⁹ we then selected four words on either side of our innovation search words without considering sentence boundaries. Lastly, we removed the duplicate words (i.e., overlaps in the data) to avoid double counting words. For example, Shutterstock's (2015) letter to shareholders described a "... quality product along with *cutting edge* tools and *technology* to empower the world's storytellers." Our innovation search words included *cutting edge* and *technology*; therefore, four words on either side of these innovation search words would be captured twice. Thus, we removed these overlaps to avoid bias from overemphasizing overlapping words.

Table 4 provides sample text illustrating the results from our collocation approach. These examples show how innovation-related terms reflect different aspects of innovation. By examining the frequency of words across all text segments, we identified *technology*, *innovation*, and *technologies* as the most common terms. Therefore, we used these words to show examples of the collocation approach. As seen in our sentence-level examples, increasing the text segment sizes may include noninnovation-related concepts or ideas.

Topic Extraction. We chose topic modeling with the Latent Dirichlet Allocation (LDA) algorithm because it is a widely applied and well-accepted topic modeling approach in management research (Blei et al., 2003; Hannigan et al., 2019). Before applying the LDA topic modeling, we undertook two initial steps: preprocessing the data¹⁰ and determining the optimal number of topics. Both of

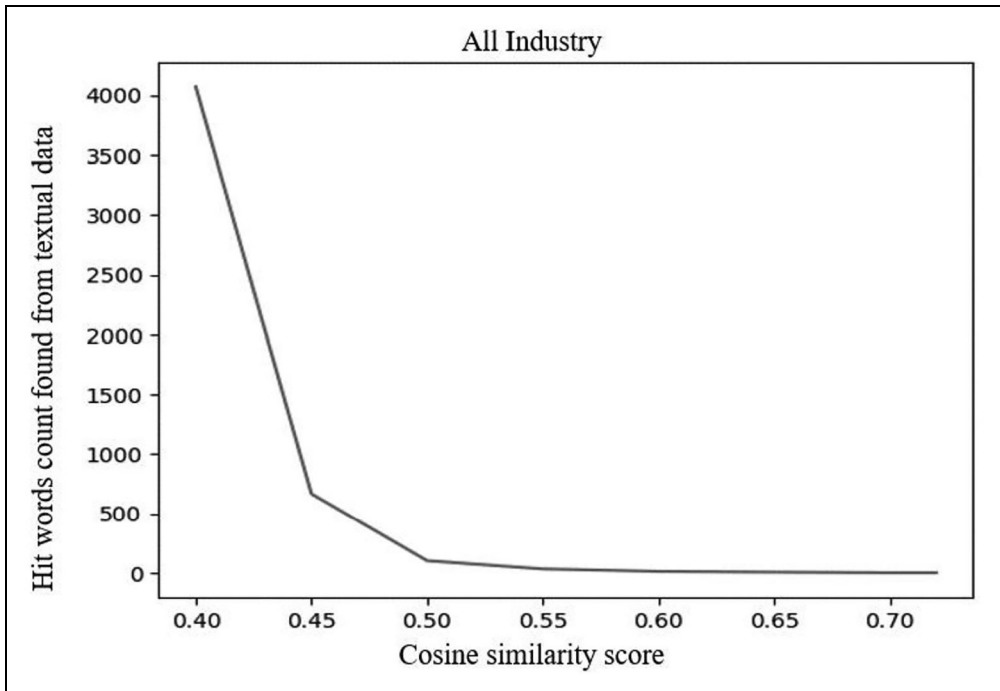


Figure 2. Cosine similarity cutoff: balancing quantity and relevance of hit words.

these processes use R code, which is provided in Appendix C along with Python code. Text-mining studies have emphasized the importance of preprocessing and suggested practical approaches for making preprocessing decisions (e.g., Hickman et al., 2022; Schmiedel et al., 2019). In our textual excerpts from the previous step, we removed stop words (which do not add value) to understand how innovation was being discussed. We also created and applied an exclusion list of words related to company-specific keywords (e.g., company and product names), industry-specific keywords (e.g., industry name and product categories), and location-specific keywords (e.g., names of countries and continents) to reduce potential noise in our topic modeling results. Additionally, we removed innovation search words before conducting the topic modeling analysis.¹¹

After the preprocessing step, researchers decide how many topics to extract. Topic modeling provides a range of outcomes, but we were looking for the optimal number of topics capturing insights into how innovation is discussed. Building on prior work (Wallach et al., 2009), we implemented a log-likelihood approach to decide the number of topics to select for further analysis. A higher log-likelihood score indicates a better model fit. However, we readily acknowledge that “there is no statistical test for the optimal number of topics or for the quality of solution” (DiMaggio et al., 2013, p. 582). As a result, we extracted multiple topics based on our log-likelihood result, compared the interpretability of each set of topics, and selected the best model after a discussion among project researchers. Figure 3 provides a scree plot of different topics’ log-likelihood output, which helped determine how many topics to extract. In this example, after extracting and evaluating 10 to 20 topics for all industries, we agreed that the relatively high log-likelihood scores (18 topics) indicated the most appropriate topics.

Each extracted topic can be reviewed as an aspect of innovation. Furthermore, researchers can combine topics into broad themes based on their understanding of the theoretical concept (Antons et al., 2016; Banks et al., 2019; Croidieu & Kim, 2018). To efficiently categorize extracted topics as broad themes, we used a quantitative approach guiding our qualitative evaluations of categorization (see Table 1, topic

Table 3. All Innovation Words From Word2Vec Process Using Google News 300.

Innovation Focal Words: Word2Vec Input	Innovation Hit Words: Word2Vec Output
Innovation, innovations, innovate, innovates, innovative, innovating.	Technological innovation, technological innovations, continually innovating, cutting edge, innovators, continually innovate, technology, technologies, innovated, continually innovates, constantly innovating, about tellabs tellabs, note ugs transforming, please visit www.networkproductsguide.com/innovations/ , intrapreneurship, ecomagination sm initiative, continuously innovating, constantly innovates, shipyard aker yards, innovator, advancements, please visit http://www.motorola.com/ , hmi scada s, copa data, please visit http://www.motorola.com , technological advancement, technological advancements, enterprise cloud orchestration, entrepreneurial mindset, voice speakerphone pendant, teamcenter tecnomatix, technologic innovation, innovation, solutionsfor, adeza designs manufactures, technological breakthroughs, pioneer pdi, technological, commercializable, stroock emphasis, ipswitch develops, cuttingedge, nortel networks enriches, suppress glucagon secretion, authors asja, spectral pattern matching, revolutionizing, biophan mri, voicegear, technol, itscustomers, adina shorr ceo, operational excellence, ita software www.itasoftware.com , ablest executives, technological advances, technological innovation, lithography optics, creativity, matrixx software, entrepreneurial spirit, technological breakthroughs, constantly striving, innovativeness, openpath, continuously innovates, entertailable, speedel develops, converged infrastructure strategy, technological developments, pioneering, invention machine innovation, inventions, continually strive, juniper networks juniper networks, customer centric, bluecherry infinity, arotech armor division, citidirect be, technologies, revolutionalize, stubhub reinvented, tyco electronics elo touchsystems, textualoud nextup.com markets, biomedexperts, productsand, jamal maraqa managing director, innovating, ###g/####g optical, narrowstep proprietary, powercloud, entrepreneurship, interdigital differentiated technology, entrepreneurial sprit, inventive, wilan founded, ulcerative colitis pediatric, excellence, called quinvaxem [®] , inovation, operationally efficient, predictive informatics.

Note: Only words or phrases that have at least a 0.5 cosine similarity score compared to the input words (i.e., focal words) are included as output words (i.e., hit words).

extraction). We used a clustering analysis (i.e., Ward’s method) to identify closely related topics (El-Hamdouchi & Willett, 1989) and then took a qualitative approach (i.e., researchers’ interpretation and discussion) to assess the clusters’ interpretability. After individually reviewing the clustered topics and labeling them, we discussed and finalized the categorization. We achieved a Krippendorff’s α of .80 as our interrater reliability and resolved one disagreement through joint discussions.

Furthermore, to ensure the interpretability of each topic and to validate its usefulness in categorizing distinct ideas, we asked four knowledgeable external scholars to interpret the topic modeling results. Instead of asking them to interpret and categorize topics, we developed survey questions for

Table 4. Data Examples After Collocation Approach.

Search Word	Four Words Before and After	Full Sentence
Technology	We build upon our technology leadership in existing markets	Looking to 2013 and beyond, we plan to accelerate our revenue growth both organically and inorganically as we build upon our technology leadership in existing markets and enter new markets in our thin films business and target new geographies with our solar energy business.
	Continue to deliver manufacturing technology that will increase productivity	We appreciate your loyalty, and we will continue to deliver manufacturing technology that will increase productivity and profitability for your business.
	Invested in our products, technology , and people in key	We strategically invested in our products, technology , and people in key areas of the business, with a focus on increasing our return on invested capital.
Innovation	Continued strong investments in innovation and sales force coverage	We expect the revenue growth across our segments, along with increasing leverage of our global infrastructure, to drive improved operating margins, even with continued strong investments in innovation and sales force coverage.
	New and existing product innovation , diversification into adjacent markets	To continue this level of growth for many years to come, we must continue to leverage our existing capabilities and strengths by investing in our business through growth in existing products, acquisitions, new and existing product innovation , diversification into adjacent markets and the aftermarket, as well as strengthening relationships with both customers and employees.
	Making investments to bring innovation to fruition that will	We are making investments to bring innovation to fruition that will expand mis surgical capabilities and continue to change the way spine surgery is being done.
Technologies	Set out to create breakthrough technologies ; we also wanted to	We didn't just set out to create breakthrough technologies ; we also wanted to improve patient lives more significantly than any other companies.
	And using our expanded technologies to explore new business	We are putting a great deal of emphasis on growth here as economic conditions continue to improve and, in the meantime, we remain focused on reducing costs and using our expanded technologies to explore new business opportunities.
	Depth of the new technologies we plan to bring	We are very excited about the breadth and depth of the new technologies we plan to bring to market which have the potential to increase our value and relevance to our customers and partners, increase differentiation versus our competitors and significantly expand our addressable market.

Note: The words “technology,” “innovation,” and “technologies” are highlighted in bold to clearly show their relation to adjacent words in our data examples.

them to complete in a process called *topic intrusion* (Chang et al., 2009; Ying et al., 2019). Chang et al. (2009) proposed this approach to evaluate topic modeling results by determining the fit between textual data and topics. We created survey questions showing each group of topics (see Table 5), which we evaluated quantitatively and qualitatively. The survey questions also included sample texts highly related to each group of topics (i.e., based on each topic's probability of occurring in the text) and a random topic that was unlikely to be related to the sample texts. For example, some sample texts might be linked to the concept of "focus on operational excellence and innovative solutions," a concept found in the fourth topic category in Table 5. We invited external participants and presented them with a group of related topics, asking them to select the least-related topic. The options included (solutions, focused, remain, driving, and leading), (operational, solutions, customers, performance, and center), and (solutions, workforce, strong, new, and focus), as well as a less-related topic: (product, new, products, market, and year). To help external participants respond to our survey questions, we defined *general innovation* and *dimensions of innovation*.

When comparing our categorizations of topic modeling's results with external participants' results, the average percentage of responses matching our categorization for the topic intrusion questions was 85%. Respondents and our author team then discussed answers differing from our initial interpretations and categorization, and we reconciled all disagreements. Thus, we believe our topics reflect semantic meaning and that topic categorizations have distinct meanings related to innovation. The clustered topic modeling results and their labels are presented in Table 5.

Next, we were curious if the groups of topics differed across high-tech and nonhigh-tech industries. As shown in Table 5, we calculated the topic distributions and found differences across industry groups, suggesting that high-tech and nonhigh-tech industries may emphasize different aspects of innovation. To assess the significant differences in discussions between the two groups, we conducted a *t*-test, seen in Table 6, which shows the average topic distributions, indicating the discussion proportions of each innovation theme across different industries.

Within the high-tech industries, we identified that one innovation-related theme—(Concept 2) significant and disruptive product innovations for customer/market opportunities—was discussed more

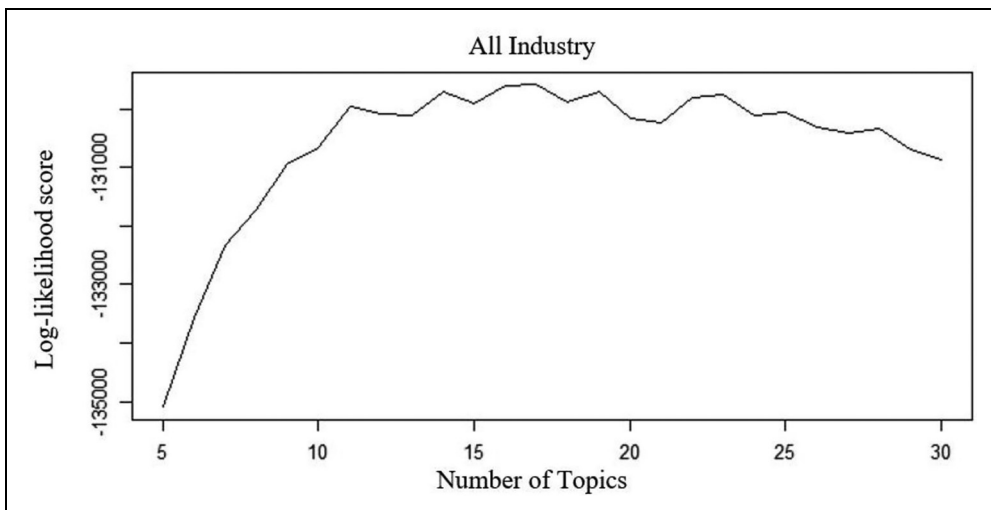


Figure 3. Log-likelihood output: identification of the optimal number of topics.

Note: A model with a higher log-likelihood score is generally considered a good fitting model. Although the highest score in all the industries can be the optimal number of topics, we extracted different numbers of topics (10 to 20 topics) to evaluate their interpretability.

than in the nonhigh-tech industries. Extracted themes—(Concept 4) focus on operational excellence and innovative solutions and (Concept 5) service excellence culture for the customer-centric innovation approach—were discussed more in the nonhigh-tech industries. We found no statistically significant industry differences in the CEOs' discussions of (Concept 1) innovation-driven growth and long-term prospects and (Concept 3) continued investments in strategic commitments (see Table 6).

These results align with prior research whereby radical innovation is a more visible factor for high-tech industries (vs. nonhigh-tech industries), given short product cycles, rapid technological changes, and quickly changing competitive landscapes (Brown & Eisenhardt, 1997; Martinez et al., 2017; Szymanski et al., 2007). Also, firms in nonhigh-tech industries are likely to focus more on operational efficiency and the development of well-established technologies in response to customer-oriented knowledge and demands (Heidenreich, 2009; Laursen & Salter, 2006; Martinez et al., 2017; Xue et al., 2012). Therefore, our WTT approach provided a detailed and nuanced understanding of how innovation is conceptualized and discussed differently across different contexts.¹²

After this point in the analysis, management researchers can use the output to examine theoretical relationships (e.g., the different topics' impact on firm performance or other outcome measures) to further develop theory (Banks et al., 2019; Kaplan & Vakili, 2015; Schmiedel et al., 2019). These topics can also be used to show changes over time if researchers have longitudinal data (Croidieu & Kim, 2018; DiMaggio et al., 2013; Zhang et al., 2021). Because this article's primary goal is to propose the WTT approach, we will not elaborate on the aforementioned possible applications.

The WTT approach necessitates several steps, decisions, and judgments similar to those in a custom dictionary or topic modeling methods, but it can provide multifaceted dimensions, nuances, and boundary conditions of the theoretical concept being studied. Therefore, we believe the WTT approach is an efficient and relevant process that can provide meaningful insights, enhancing theory development in emerging or less-explored areas of management research.

Comparison With Traditional Approaches

Before implementing the WTT approach, we compared our results with results using the innovativeness dictionary (Short et al., 2010) and LDA topic modeling (Blei et al., 2003) using our corpus of CEO letters to shareholders.

First, after applying the innovativeness dictionary to our full documents, we found that the dictionary words captured only a small amount of the total corpus (7,519 out of 920,446 words or 0.82%). This finding indicates that while the dictionary captured the innovativeness construct (i.e., a general sense of innovation), nearly 99% of the remaining text (i.e., leftover words) was unexplored. Thus, to fully explore the innovation concept within the corpus, additional examination of these leftover words was necessary. However, building different types of innovation dictionaries, reviewing words through a KWIC, or conducting manual coding is extremely time-consuming with 912,927 remaining words.

Second, we conducted LDA topic modeling using letters to the shareholders (Blei et al., 2003). We extracted 40 topics using all our documents and found various business-related topics related, such as global expansion, business strategy, financial performance, and top management team, among others. However, innovation-related topics were not obvious in this extraction using all the text in our documents (see Appendix E).

Discussion

In this study, we compared various text analysis approaches and proposed our WTT approach. In applying WTT to innovation, we identified innovation search words (i.e., focal and hit words) through word extraction, focused on meaningful parts of the text associated with our theoretical

Table 5. LDA Topic Modeling Results of High-Tech and Nonhigh-Tech Industries (After Researchers' Labeling and Categorization).

Researcher Aggregated Concepts	Topics	Topic Distribution*		Extracted Words in Topics
		High-Tech	NonHigh-Tech	
(1) Innovation-driven growth and long-term prospects	Topic 1	0.047	0.036	New, award, design, ability, growing, key, network, enhance, first, center
	Topic 4	0.054	0.043	Growth, new, data, portfolio, investment, unique, customer, flow, investing, year
	Topic 11	0.050	0.048	New, create, can, many, future, breakthrough, just, system, believe, pipeline
(2) Significant and disruptive product innovations for customer/market opportunities	Topic 5	0.074	0.047	Advanced, new, products, leadership, capabilities, continued, customers, engineering, sales, well
	Topic 7	0.052	0.065	Product, new, manufacturing, quality, development, leadership, deliver, design, develop, customers
	Topic 13	0.061	0.033	Products, proprietary, acquired, applications, core, key, based, platform, develop, disruptive
	Topic 15	0.079	0.054	Product, new, products, market, year, growth, company, significant, development, opportunities
	Topic 18	0.062	0.029	Control, process, products, markets, solutions, new, next, acquisitions, generation, device
(3) Continued investments in strategic commitments	Topic 8	0.050	0.061	Operational, market, platform, continue, delivery, service, strategic, digital, position, focus
	Topic 10	0.052	0.039	Market, investments, record, focus, track, focused, strong, teams, platform, speed
	Topic 16	0.066	0.050	Continued, growth, drive, continue, commitment, success, across, dedication, investments, since
(4) Focus on operational excellence and innovative solutions	Topic 9	0.062	0.055	Solutions, focused, remain, driving, leading, delivering, global, company, commitment, committed
	Topic 12	0.040	0.085	Operational, solutions, customers, performance, center, leadership, growth, development, provider, focus
	Topic 17	0.037	0.068	Solutions, workforce, strong, new,

(continued)

Table 5. (continued)

Researcher Aggregated Concepts	Topics	Topic Distribution*		Extracted Words in Topics
		High-Tech	NonHigh-Tech	
(5) Service excellence culture for customer-centric innovation approach	Topic 2	0.058	0.069	focus, position, products, processes, customer, developing Expertise, focus, work, culture, best, people, marketing, business, awards, believe
	Topic 3	0.056	0.078	Services, new, solutions, clients, deliver, customers, communications, financial, value, data
	Topic 6	0.056	0.050	Industry, service, quality, customer, integrity, products, across, customers, values, spirit
	Topic 14	0.044	0.090	Information, services, business, systems, management, capabilities, officer, strategy, chief, solutions

Note: An asterisk (*) indicates the topic distribution which is the probability of a topic's distribution across all documents. It shows us which topics are the most and least frequent in the corpus, and the sum of all scores is 1.

Table 6. Comparison of Topic by Industry Type.

Researcher Aggregated Concepts	Number of Topics	Average Topic Distribution by Industry			t-Test Statistic	p-value
		Total	High-Tech	Nonhigh-Tech		
(1) Innovation-driven growth and long-term prospects	3	0.139	0.151	0.126	1.637	.102
(2) Significant and disruptive product innovations for customer/market opportunities	5	0.278	0.329	0.229	4.714	.001
(3) Continued investments in strategic commitments	3	0.159	0.168	0.150	1.038	.299
(4) Focus on operational excellence and innovative solutions	3	0.174	0.139	0.208	-4.019	.001
(5) Service excellence culture for customer-centric innovation approach	4	0.250	0.213	0.287	-3.720	.001

Note: (1) A two-sample t-test was conducted to compare the groups.

(2) Although we initially started with 841 letters, after the collocation step, some letters that did not contain our innovation search words were excluded. This reduced our sample size to 719 letters.

concept of interest through a collocation process, and extracted topics from the text segments through topic modeling. With this WTT approach, we identified different aspects of innovation and found that they varied across industry contexts. In high-tech industries, we found more discussion of disruptive

innovation; whereas in nonhigh-tech industries, we found more discussion of customer-centered innovations and operational excellence. We believe our approach contributes to a process to further theory development by suggesting a way to capture aspects of a theoretical area of interest.

Moreover, we believe the WTT approach has a wide range of potential applications in management research. Given the vast amount of textual data available to management researchers today (e.g., internal company documents, news media, and social media), text analysis holds great promise to further understanding of a theoretical domain. Although some skill is needed to undertake our process, we used open-source programs (Python and R), which are widely available; and we have provided step-by-step instructions to guide novice researchers through this process (see Appendices A, B, and C).

Our study, however, highlights some limitations that require further study and consideration. First, our WTT approach does not focus on construct measurement through rigorous validity processes, which is a strength of the custom dictionary approach (Short et al., 2010). In other words, the WTT approach does not capture construct measurement for a specific theoretical concept. Instead, its outcomes share similarities with topic modeling approaches, which are more exploratory.

Second, another concern is that by applying a collocation approach, salient information related to the theoretical domain will be lost. As seen in our example, most of the text in the CEOs' letters to shareholders was discarded after our collocation process. While this approach reduces the noise of capturing irrelevant information and focuses on the selected construct in the text, it also removes available data to which topic modeling could be applied. However, when we conducted LDA topic modeling with the full data, we obtained over 40 topics with few having face validity related to an innovation construct. Thus, future studies should clearly describe how they managed this balancing issue—minimizing irrelevant information while capturing topics related to a theoretical domain of interest.

Third, there is a lack of guidance on a collocation approach, specifically the number of words to select around the search words (i.e., focal and hit words). We urge future researchers to compare and report results using text segments of varying lengths. In our study, there were no significant differences in topics when we included four or five words on either side of our innovation focal and hit words; therefore, we selected the more parsimonious solution. When we included more words on either side (e.g., up to 10 on each side) or used entire sentences in which a search word was found, we discovered that many of the resulting topics did not have an obvious connection to the innovation concept. In future studies using this WTT approach, researchers should consider assessing how different text segments around the search words affect the topics created and the resulting interpretation.

Additionally, with advancements in natural language processing (NLP) techniques, more sophisticated approaches can be used with our WTT approach. In the topic extraction step, there are advanced techniques (e.g., BERTopic and Top2Vec), which can offer novel alternatives. By using embedding and clustering techniques before topic extraction, these techniques have the potential to generate contextually related topics and make their interpretation easier (Angelov, 2020; Grootendorst, 2022). These approaches may provide more nuanced representations of textual data and improve extracted topics' relevance and interpretability. In addition, researchers can also use n-gram, sequences of words (i.e., phrases), as an alternative option to single-word topic modeling. This approach may help researchers capture more context and enrich topics by including meaningful phrases, thus enhancing the understanding of a theoretical domain.

This article's purpose is to develop an understanding of the WTT approach by applying it in the innovation theoretical domain. Concerns with our approach and new NLP techniques necessitate more investigating, reporting, and refining as researchers use it to explore a wide variety of constructs and theoretical domains in management research. Despite these limitations, our experience reveals a way forward for management researchers to efficiently and effectively explore a theoretical domain

in large textual datasets. We believe that this article will help researchers decide which text analysis approach to select and which provides another option in their decision-making.

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Supplemental Material

Supplemental material for this article is available online.

Notes

1. The CAT scanner website (<http://www.catscanner.net/dictionaries/>), created by well-known computer-aided text analysis (CATA) researchers Aaron McKenny and Jeremy Short, provides access to numerous custom dictionaries that have been published in peer-reviewed management and organizational journals.
2. Researchers may choose their preprocessing strategy based on their specific needs. For example, if a researcher wants to capture sentence-level (i.e., full sentences containing a search word) rather than word-level collocation (i.e., a certain number of words *before* and *after* a search word), they do not have to remove punctuation at this stage. A sentence tokenizer might be useful for identifying sentence boundaries. If ensuring the letter case of proper nouns (e.g., names of people, places, and organizations) is important, named entity recognition can be used to find these phrases and recode them before searching for the collocation patterns around the search words (Nadeau & Sekine, 2007).
3. To explain the data more specifically, high-tech industries include firms from computer equipment (SIC 357), electrical (SIC 367), medical technology (SIC 382, 384), and software (SIC 737). Nonhigh-tech industries include firms from machinery and engines (SIC 355, 356), motor vehicles (SIC 371), and management services (SIC 874).
4. We were unable to collect all CEOs' letters to shareholders because some firms did not publish those letters in certain years or the letters were not available in our databases (firm's webpage, annualreports.com, and Mergent database).
5. We acknowledge that our corpus may not contain content and words similar to those in the Google word2vec model trained with news articles. To address this issue, we attempted to train a custom word2vec model, hoping to capture innovation-related hit words from our corpus. However, due to the limited size (841 documents) of our textual data, we were unable to train an adequate model. Therefore, we obtained many hit words that were unrelated to our focal innovation words. Given this issue, we used only the pretrained model in this study.
6. The similarity score between two words is calculated by using vectors of words that indicate semantic similarities (Bolukbasi et al., 2016). The score's formula presents as $\cos(u, v) = \frac{u \cdot v}{|u||v|}$. In this formula, u is the word vector for focal words and v is Google's word vectors (or word vectors from CEOs' letters to shareholders). Using this formula, researchers can capture the distance between the focal words and words from textual data (Google or CEOs' letters to shareholders) and obtain the focal words' hit words.

7. We also reviewed other text segment sizes (e.g., 10 words on each side, sentence levels, and paragraph levels) and decided four words on each side of the search words because longer text segments contained more noninnovation-related information.
8. Although spelled the same, uppercase and lowercase words are treated differently in such programming languages as R and Python. To correctly identify words regardless of case, researchers may need to convert all text to the same case.
9. Text extraction can be accomplished in either R with the KWIC function or Python with the concordance function.
10. The preprocessing task in the text extraction was conducted to easily find matches of the search words and capture clear concept-related textual units (e.g., paragraphs, sentences, or phrases) for exploration. Heavily preprocessed textual data (e.g., stemming) may make it difficult to find the right matches. The preprocessing task in the topic extraction identified clear themes related to the concept. To improve the topics' interpretability by further removing noise or errors, more extensive preprocessing efforts may be necessary.
11. This was done because the presence of the search words in each text segment could overly emphasize them in the analysis, potentially resulting in extracting topics mainly related to the search words, rather than uncovering other relevant words within the extracted text segments.
12. We also applied our WTT approach to datasets from the high-tech and nonhigh-tech industries separately (see Appendix D). The results aligned with our earlier findings when analyzing from all industry data. Specifically, we noticed similar topics were emphasized within each industry, such as disruptive innovations for high-tech industries and incremental/adaptive innovation to fit the evolving needs of customers for nonhigh-tech industries.

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